

Introduction to Databases Project

Author: [Massimiliano Mola](#) Student Nr. 23026

Completed on 16.02.2025

The project aims to be a simplified version of the The Hitchhiker's Guide to the Galaxy

The project was build starting from prof. Cavalese's [project requirements](#).

Specification: The HHGTTG Repository

We are interested in the Guide that contains entries helpful for navigating the universe, such entries have to be divided in the entries regarding VIPs (Arthur Dent: A confused Earthman who found himself rather unexpectedly thrust into galactic adventures. Entry notes: "Mostly in search of tea and a decent sandwich."), entries regarding planets ("Earth: Mostly harmless. Except for the moments when it isn't. Known for its bizarre obsession with paperwork and reality TV."), entries regarding species (Vogon Poetry: The third worst poetry in the universe. Exposure to it can cause extreme nausea, loss of will to live, and in extreme cases, spontaneous self-combustion) and general entries (Towel: The single most massively useful thing an interstellar hitchhiker can carry. It can be used for warmth, defense, signaling, or even as a makeshift flotation device. Most importantly, it makes you look like you know what you're doing). Only approved Authors with a high enough reputation, may add entries to the Guide. Authors. VIPs, who are generally famous persons and not only authors may be given a score, to quickly identify how important they are eg. the president of the galaxy should have a high score. In addition, we are interested in the travelers using our guide, which may rate the location they visited on a scale from 0 to 100, can rate a location more than once provided at least one 30 standard days passed since the last visit. We need also the time We are also interested in the Spacecraft they use to travel, in particular how many people it transported, the amenities of the vehicle, and the name of the organizations that produce such spacecraft. We want to know the planet of origin of the travelers, the species they belong to and which planets they visited,

Structured and organized requirements

1. Guide Entries Management

- The system must manage a repository of entries that provide useful information for navigating the universe.
- The system must classify each entry into one of the following categories:
 - VIP entries (for important or famous persons)
 - Planet entries
 - Species entries
 - General entries

2. Authorship and Approval

- The system must restrict the ability to add new entries to approved authors only.
- Approval must be based on criteria ensuring the author's suitability to contribute to the Guide.

3. **VIPs**

- The system must identify VIPs as persons who hold a notable status (e.g., public figures, high-ranking officials, etc.).
- The system must allow for a measure of importance or significance to be assigned to VIPs.

4. **Travelers and Visits**

- The system must keep track of travelers who make use of the Guide.
- The system must record each traveler's visits to various locations across the universe.
- The system must record the planet of origin for each traveler.

5. **Ratings and Constraints**

- The system must allow travelers to evaluate or rate a location after visiting it (from 0 to 100).
- The system must ensure that a traveler can submit multiple ratings for the same location only if certain temporal conditions are met.

6. **Spacecraft**

- The system must keep track of the spacecraft used by travelers for their journeys.
- The system must record details about the organizations responsible for producing these spacecraft.

7. **Species and Origin**

- The system must record the species to which each traveler belongs.
- The system must track the planet of origin for each traveler.

8. **Planetary Information**

- The system must maintain information about the planets in the universe.
- The system must track which travelers have visited which planets.

9. **Temporal Tracking**

- The system must handle the notion of time and ensure time-related rules (e.g., intervals between ratings) are enforced.

common operations

Below are five of the most common operations that the system is expected to perform:

1. **Adding a New Guide Entry**

- Verify that the contributor is an approved author.
- Insert a new entry into the Entry table.

- Classify the entry by inserting a corresponding record into one of the classification tables (PersonEntry, LocationEntry, or SpeciesEntry).

2. Recording a Traveler's Trip

- insert a new record into the Trip table to represent a traveler's journey.

3. query the most visited location in a sector

4. Managing Spacecraft Usage

- Track which spacecraft a traveler uses during a trip.
- Capture spacecraft details such as name, capacity, amenities, and the associated manufacturing organisation.

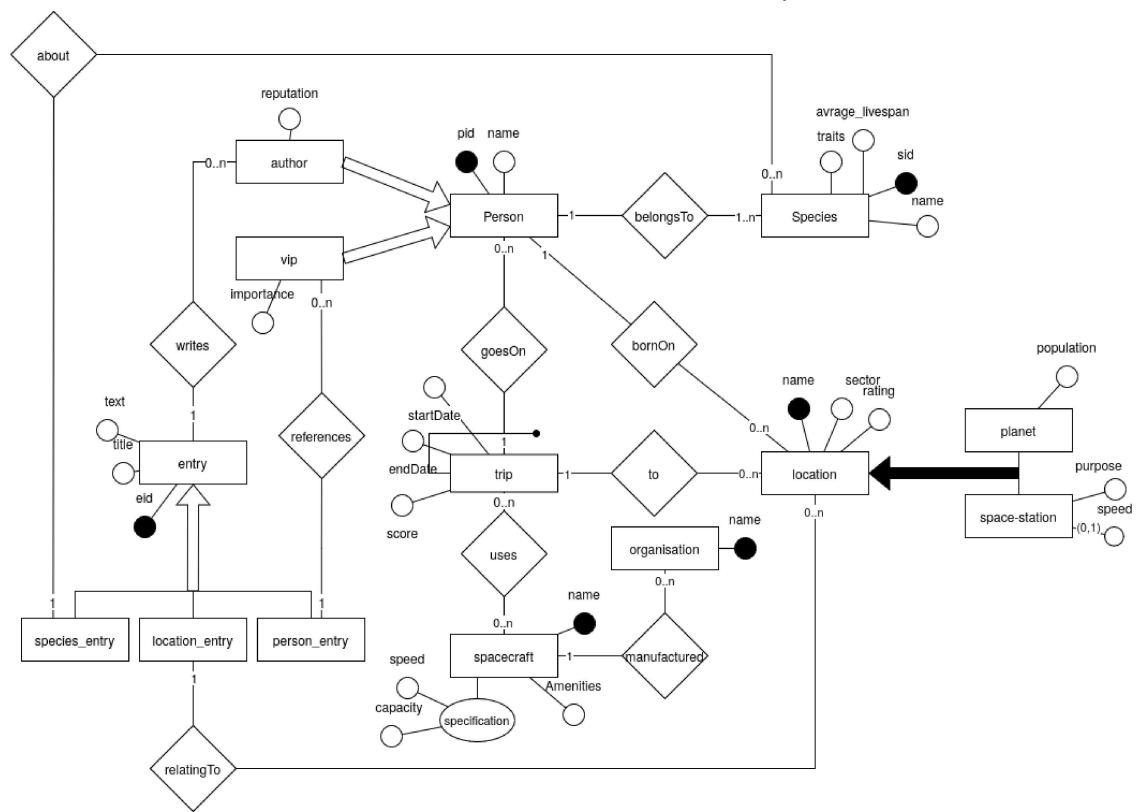
5. Querying and Entries about a location

- Retrieve entries based on various criteria (e.g., by category: VIP, location, species, or general).
- Enable filtering of entries by attributes like ratings, author, or related location.

glossary

term	description	synonym	connections
person	persons travel, write articles and/or are VIPs	individual	species, location, spacecraft
location	planets, stars, space stations	place	person, organisation
species	species of persons	race	person, location
organisation	organisations that produce spacecraft	company	spacecraft
spacecraft	spacecraft used by persons	ship	person, organisation, location
entry	entries in the guide	article	author
author	authors of entries	writer	entry, person
vip	very important person	important person	person, person_entry
person_entry	rating of a person	rating	vip, entry
location_entry	rating of a location	rating	location, entry
species_entry	rating of a species	rating	species, entry
planet	planets in the universe	world	location
space_station	space stations in the universe	station	location

diagram of the conceptual schema



data dictionary

Entity	Description	Attributes	Identifier(s)
Entry	A guide entry that provides useful information for navigating the universe.	ied, title, text	ied
Person	An individual in the system (e.g., traveler, author, VIP).	pid, name	pid
Location	A physical location in the universe (can be a planet or a space station).	name, sector, rating	name
Organisation	An organization responsible for producing spacecraft.	name	name
Spacecraft	A vehicle used by travelers for journeys.	name, capacity, amenities, speed	name
Species	A biological species in the universe.	sid, name, traits, average_lifespan	sid
Trip	A travel event representing a journey with an evaluation score and duration.	startDate, endDate, person, score	Composite: (startDate, endDate, person)

Entity	Description	Attributes	Identifier(s)	
Author	An approved contributor who can add new entries to the Guide.	id, reputation	id	
Vip	A person with notable status (e.g., public figure, high-ranking official).	id, importance	id	
PersonEntry	Marks an entry as a person-related entry.	entry	entry	
LocationEntry	Marks an entry as a location-related entry.	entry	entry	
SpeciesEntry	Marks an entry as a species-related entry.	entry	entry	
Planet	Information about a planet; a subset of Location.	name, population	name	
SpaceStation	Information about a space station; a subset of Location.	name, purpose, speed*	name	
Relationship	Description	Components	Attributes	Identifier(s)
about	Associates an entry with the species it is about.	entry, species	None	implicit
writes	Indicates that an author wrote a given entry.	entry, author	None	implicit
references	Indicates that an entry references a VIP.	entry, vip	None	implicit
relatingTo	Associates an entry with a location it is related to.	entry, location	None	implicit
belongsTo	Indicates that a person belongs to a specific species.	person, species	None	implicit
goesOn	Indicates that a person embarks on a trip.	person, startDate, endDate	None	implicit

Relationship	Description	Components	Attributes	Identifier(s)
uses	Indicates that a person uses a particular spacecraft during a trip.	person, spacecraft, startDate, endDate	None	implicit
to	Indicates that a person travels to a specific location as part of a trip.	person, location, startDate, endDate	None	implicit
bornOn	Specifies the planet on which a person was born.	person, planet	None	implicit
manufactured	Associates a spacecraft with the organisation that manufactured it.	spacecraft, organisation	None	implicit

table of volumes and table of operations according to the foreseen application load

table of volumes

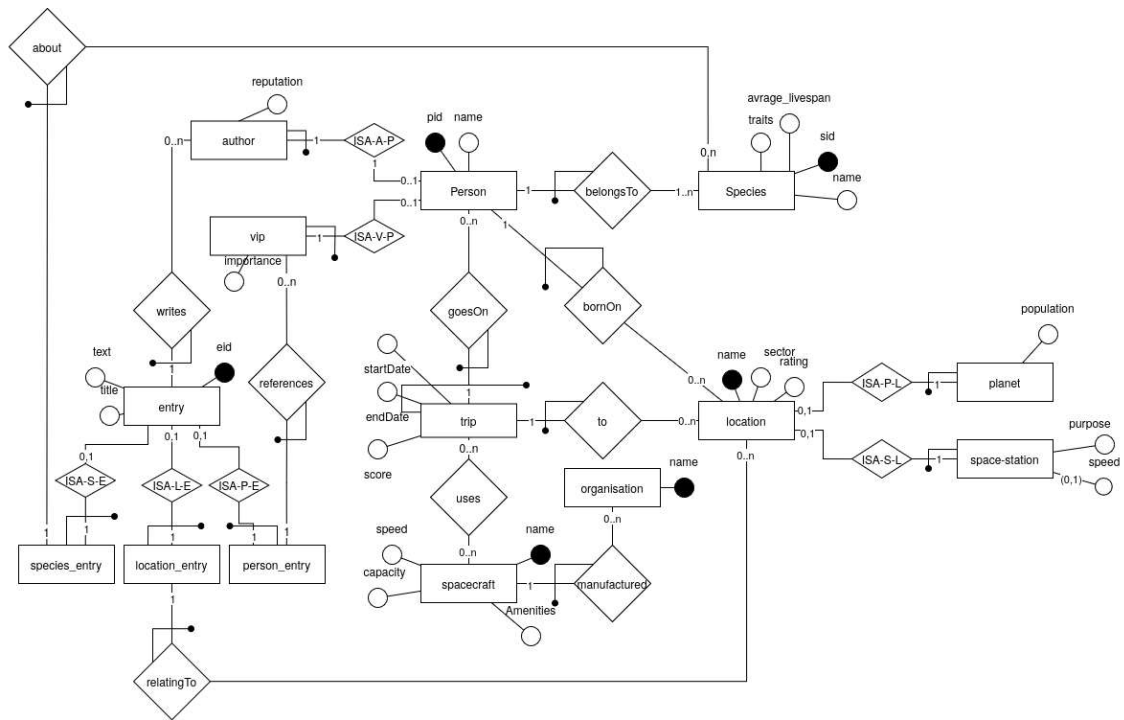
Concept	Construct	Volume
Entry	Entity	1,000,000
Person	Entity	100,000
Location	Entity	10,000
Organisation	Entity	100
Spacecraft	Entity	1,000
Species	Entity	1,000
Trip	Entity	1,000,000
Author	Entity	1,000
Vip	Entity	10,000
PersonEntry	Entity	10,000
LocationEntry	Entity	100,000
SpeciesEntry	Entity	10,000
Planet	Entity	5,000
SpaceStation	Entity	5,000

Concept	Construct	Volume
writes	Relationship	1,000,000
about	Relationship	10,000
references	Relationship	10,000
relatingTo	Relationship	100,000
belongsTo	Relationship	100,000
goesOn	Relationship	1,000,000
uses	Relationship	1,000,000
to	Relationship	1,000,000
bornOn	Relationship	100,000
manufactured	Relationship	1,000

table of operations

Operation	Description	Frequency
Add New Entry	Add a new guide entry	1/day
Record Traveler's Trip	Record a traveler's trip	100/day
Submit Location Rating	Allow a traveler to rate a location	10/day
Manage Spacecraft Usage	Track spacecraft usage	1/day
Query Guide Information	Retrieve guide entries	1,000/day

Restructured conceptual scema



external constraints

1. Each instance of Entry participates to at most one of the relationships ISA-L-E or ISA-P-E or ISA-S-E
2. Each instance of Location participates to exactly one of the relationships ISA-S-L, ISA-P-L
3. In each instance of Trip startDate must be earlier than endDate
4. the rating in Trip and location must be between 0 and 100

translation

Entry(ied, title, text)

Person(pid, name)

Location(name, sector, rating) Organisation(name)

Spacecraft(name, capacity, amenities, speed)

Species(sid, name, traits, average_lifespan)

Trip(startDate, endDate, person, score)

Foreign key: Trip[person] $\subseteq$ Person[pid]

Foreign key: Trip[location] $\subseteq$ Location[name]

Author(id, reputation)

Foreign key: Author[id] $\subseteq$ Person[pid]

Vip(id, importance) Foreign key: Vip[id] $\subseteq$ Person[pid]

PersonEntry(entry)

Foreign key: PersonEntry[entry] $\subseteq$ Entry[ied]

LocationEntry(entry)

Foreign key: LocationEntry[entry] $\subseteq$ Entry[id]

SpeciesEntry(entry)

Foreign key: SpeciesEntry[entry] $\subseteq$ Entry[id]

Planet(name, population)

Foreign key: Planet[name] $\subseteq$ Location[name]

SpaceStation(name, purpose, speed*)

Foreign key: SpaceStation[name] $\subseteq$ Location[name]

about(entry, species)

Foreign key: about[entry] $\subseteq$ Entry[id]

Foreign key: about[species] $\subseteq$ Species[sid] \

writes(entry, author)

Foreign key: writes[entry] $\subseteq$ Entry[id]

Foreign key: writes[author] $\subseteq$ Author[id]

references(entry, vip)

Foreign key: references[entry] $\subseteq$ Entry[id]

Foreign key: references[vip] $\subseteq$ Vip[id]

relatingTo(entry, location)

Foreign key: relatingTo[entry] $\subseteq$ Entry[id]

Foreign key: relatingTo[location] $\subseteq$ Location[name]

belongsTo(person, species)

Foreign key: belongsTo[person] $\subseteq$ Person[pid]

Foreign key: belongsTo[species] $\subseteq$ Species[sid]

goesOn(person, startDate, endDate)

Foreign key: goesOn[person] $\subseteq$ Person[pid]

Foreign key: goesOn[startDate] $\subseteq$ Trip[startDate]

Foreign key: goesOn[endDate] $\subseteq$ Trip[endDate]

uses(person, spacecraft, startDate, endDate)

Foreign key: uses[person] $\subseteq$ Trip[person]

Foreign key: uses[spacecraft] $\subseteq$ Spacecraft[name]

Foreign key: uses[startDate] $\subseteq$ Trip[startDate]

Foreign key: uses[endDate] $\subseteq$ Trip[endDate]

to(person, location, startDate, endDate)

Foreign key: to[person] $\subseteq$ Trip[person]

Foreign key: to[location] $\subseteq$ Trip[startDate]

Foreign key: to[endDate] $\subseteq$ Trip[endDate]

bornOn(person, planet)

Foreign key: bornOn[person] $\subseteq$ Person[pid]

Foreign key: bornOn[planet] $\subseteq$ Planet[name]

manufactured(spacecraft, organisation)

Foreign key: manufactured[spacecraft] $\subseteq$ Spacecraft[name]

Foreign key: manufactured[organisation] $\subseteq$ Organisation[name]

external constraints

- Spacestation[name] $\cap$ Planet[name] = $\emptyset$
- Location[name] $\subseteq$ Planet[name] $\cup$ Spacestation[name]
- LocationEntry[entry] $\cap$ SpeciesEntry[entry] $\cap$ PersonEntry[entry] = $\emptyset$
- Trip[startDate] < Trip[endDate]
- the rating in Trip and location must be between 0 and 100

restructuring of the relation schema

We merge most (1,1) to (0/1,n) relationships into the entity they are connected to reduce the complexity of the schema.

Entry(ied, title, text, author)

FK: Entry[author] $\subseteq$ Author[id]

Person(pid, name, species, bornon)

FK: Person[species] $\subseteq$ Species[sid], Person[bornon] $\subseteq$ Location[name]

Location(name, sector, rating)

Organisation(name)

Spacecraft(name, capacity, amenities, organisation) FK: Spacecraft[organisation] $\subseteq$ Organisation[name]

Species(sid, name, traits, avrage_lifespan)

Trip(startDate, endDate, person, location, score)

Foreign key: Trip[person] $\subseteq$ Person[pid]

Foreign key: Trip[location] $\subseteq$ Location[name]

Author(id, reputation)

Foreign key: Author[id] $\subseteq$ Person[pid]

Vip(id, importance) Foreign key: Vip[id] $\subseteq$ Person[pid]

PersonEntry(entry, vip)

Foreign key: PersonEntry[entry] $\subseteq$ Entry[ied], PersonEntry[vip] $\subseteq$ Vip[id]

LocationEntry(entry, location)

Foreign key: LocationEntry[entry] $\subseteq$ Entry[ied], LocationEntry[location] $\subseteq$ Location[name]

SpeciesEntry(entry, species)

Foreign key: SpeciesEntry[entry] $\subseteq$ Entry[ied], SpeciesEntry[species] $\subseteq$ Species[sid]

Planet(name, population)

Foreign key: Planet[name] $\subseteq$ Location[name]

SpaceStation(name, purpose, speed*)

Foreign key: SpaceStation[name] $\subseteq$ Location[name]

uses(person, spacecraft, startDate, endDate)

Foreign key: uses[person] $\subseteq$ Trip[person]

Foreign key: uses[spacecraft] $\subseteq$ Spacecraft[name]

Foreign key: uses[startDate] $\subseteq$ Trip[startDate]

Foreign key: uses[endDate] $\subseteq$ Trip[endDate]

external constraints

- Spacestation[name] $\cap$ Planet[name] = $\emptyset$
- Location[name] $\subseteq$ Planet[name] $\cup$ Spacestation[name]
- LocationEntry[entry] $\cap$ SpeciesEntry[entry] $\cap$ PersonEntry[entry] = $\emptyset$
- Trip[startDate] < Trip[endDate]
- the rating in Trip and location must be between 0 and 100